# PoseInsight: A Hybrid AI System for Real-Time Exercise Form Classification, Movement Quality Assessment, and Intelligent Coaching

Satyabrata Das
Applied Deep Learning
University of Florida
Email: satyabradas@ufl.edu
Instructor: Andrea Ramirez-Salgado
University of Florida
Email: aramirezsalgado@ufl.edu

***Abstract*—Human movement analysis is critical in fitness coaching, rehabilitation, sports science, and injury prevention. While pose estimation technologies have made real-time body landmark detection increasingly accessible, many existing systems remain limited to skeleton visualization or repetition counting without deeper biomechanical understanding. This paper presents PoseInsight, a hybrid AI-powered movement intelligence system designed to bridge the gap between raw pose estimation and actionable exercise-quality assessment. PoseInsight integrates MediaPipe-based pose estimation, structured biomechanical feature engineering, comparative machine learning classification, and interactive deployment supporting both uploaded video and real-time webcam input. The system extracts 240 interpretable movement features per frame, including joint angles, posture alignment metrics, bilateral symmetry ratios, and sliding window motion statistics, enabling classification of exercise type and movement quality across four categories: good squat, bad squat, good push-up, and bad push-up. Multiple modeling approaches were explored and systematically compared, including a rule-based finite state machine baseline, a Multi-Layer Perceptron (MLP), a Long Short-Term Memory network (LSTM), and a Random Forest classifier. Experimental evaluation on a custom 168-sample labeled dataset demonstrated that Random Forest achieved the strongest validation accuracy of 97%, outperforming MLP (85%) and LSTM (88%) under constrained dataset conditions. These findings demonstrate that carefully engineered biomechanical features combined with robust classical machine learning can outperform more complex temporal deep learning architectures when labeled data is limited. PoseInsight establishes a practical foundation for future intelligent coaching systems incorporating personalized feedback, transformer-based temporal modeling, and clinical biomechanical reasoning.**

***Index Terms*—Human Pose Estimation, Exercise Classification, Random Forest, LSTM, Biomechanics, Computer Vision, Explainable AI, Intelligent Coaching**

## I. Introduction

Human movement quality directly impacts athletic performance, rehabilitation success, and injury prevention. Improper exercise execution can lead to poor outcomes, chronic injuries, or ineffective training. Traditionally, movement analysis has relied heavily on in-person coaching, biomechanical laboratories, or subjective self-assessment, all of which can be expensive, inaccessible, or inconsistent.

Research indicates that approximately 70% of recreational athletes experience at least one exercise-related injury annually, many of which are directly attributable to poor movement mechanics and the absence of real-time corrective feedback [7]. Traditional coaching models suffer from an inherent delay problem: by the time a coach observes and verbally corrects a movement error, the repetition has already been completed incorrectly, reinforcing harmful motor patterns. Real-time automated feedback systems can interrupt this cycle by providing instantaneous, objective movement quality assessments at the moment of execution, enabling immediate correction before poor form becomes habitual.

Recent advances in computer vision have democratized access to pose estimation technologies capable of extracting body landmarks from consumer-grade cameras. Frameworks such as MediaPipe Pose allow real-time body landmark detection without specialized equipment. However, despite these advancements, many practical implementations remain limited to basic skeleton rendering or repetition counting.

This limitation creates a major gap: pose detection alone does not constitute movement intelligence. True intelligent movement systems must evaluate quality, posture, biomechanical correctness, and potential injury risks.

PoseInsight addresses this gap by transforming pose estimation into a hybrid AI-powered movement intelligence system capable of:

- Exercise classification
- Good vs bad movement quality analysis
- Biomechanical feature extraction
- Comparative machine learning evaluation
- Real-time deployment

Unlike purely threshold-based systems, PoseInsight evolved from an initial rule-based prototype into a comparative machine learning framework involving MLP, LSTM, and Random Forest models.

The primary contributions of this work include:

1) Development of an end-to-end hybrid AI pipeline integrating perception, feature engineering, and deployment.
2) Construction of a labeled exercise dataset containing good and bad squat/push-up examples.
3) Comparative evaluation of classical ML and deep learning approaches.
4) Demonstration that Random Forest outperformed LSTM and MLP under limited biomechanical dataset conditions.
5) Establishment of a practical foundation for future intelligent coaching and movement assessment systems.

This work emphasizes a key practical insight: model complexity alone does not guarantee superior performance—effective feature engineering and dataset alignment are equally critical.

It is worth emphasizing that PoseInsight was developed under realistic resource constraints representative of independent research settings: no motion capture laboratory, no force plates, no wearable sensors, and no large pre-existing labeled dataset. The system relies entirely on a consumer webcam, open-source libraries, and a custom-collected video dataset. This constraint-driven design philosophy is intentional — demonstrating that meaningful movement intelligence is achievable without specialized infrastructure is itself a contribution to the democratization of biomechanical assessment tools.

The complete source code, training notebooks, and Streamlit deployment application for PoseInsight are publicly available to support replication and further research:

https://github.com/Satyabratadas/pose-insight

The remainder of this paper is organized as follows: Section II reviews related work and compares PoseInsight to existing approaches. Section III describes the system design and AI lifecycle architecture. Section IV details dataset collection and preprocessing. Section V presents the model development process across rule-based, MLP, LSTM, and Random Forest approaches. Section VI reports evaluation results and comparative analysis. Section VII discusses key findings, strengths, and limitations. Section VIII addresses responsible AI considerations. Section IX outlines future work directions, followed by the conclusion in Section X.

## II. Related Work

Pose estimation has rapidly advanced through lightweight systems such as BlazePose [1], which provides real-time body landmark extraction optimized for fitness and rehabilitation contexts. Bazarevsky et al. demonstrated that BlazePose achieves competitive landmark localization accuracy on consumer hardware by using a two-stage detector-tracker pipeline, enabling 30+ FPS inference on mobile devices. While BlazePose effectively detects 33 body landmarks including fine-grained hand and foot points, it outputs raw coordinate data without any interpretation of movement quality, leaving the clinical or coaching assessment entirely to downstream systems.

Research in exercise recognition has explored rule-based systems using joint angle thresholds and finite state machines [6]. Velloso et al. collected the widely-used Weight Lifting Exercises dataset, in which participants performed bicep curls in five distinct quality categories while wearing inertial measurement units. Their rule-based classification approach achieved strong within-subject accuracy but demonstrated significant performance degradation across subjects with different body proportions and movement styles. These findings highlight the fundamental limitation of purely threshold-based systems: fixed angle thresholds cannot generalize across the natural variability of human movement without extensive per-user calibration.

Human Activity Recognition (HAR) literature has increasingly adopted machine learning and deep learning models to overcome the generalization limitations of rule-based approaches [7]. Bulling et al. provided a comprehensive survey of body-worn sensor-based HAR systems, demonstrating that feature engineering remains critical even for deep learning pipelines. Their work established that model performance is tightly coupled to the quality and relevance of input representations, not solely to model architecture complexity. However, most HAR benchmarks focus on activity-type recognition rather than within-activity quality assessment, leaving a gap that PoseInsight directly addresses.

Biomechanical literature has established the importance of joint angles for movement assessment in clinical and athletic contexts. Escamilla et al. conducted a comprehensive biomechanical analysis of squat mechanics, demonstrating that knee flexion angle, hip angle, and trunk lean are the primary determinants of both performance quality and injury risk during loaded squat movements [3]. Their findings provide direct biomechanical justification for the joint angle features central to PoseInsight's feature engineering pipeline. Similarly, Suprak et al. analyzed push-up biomechanics, establishing that elbow flexion range and body alignment are the critical quality indicators for push-up execution [4], both of which are explicitly captured in PoseInsight's extracted feature set.

Compared with prior work, PoseInsight uniquely addresses the intersection of real-time pose estimation, biomechanical feature engineering, multi-class movement quality classification, and interactive deployment within a single unified system. Unlike BlazePose-only implementations, PoseInsight adds a full intelligence layer above landmark detection. Unlike HAR systems that classify activity types, PoseInsight classifies movement quality within activity types. Unlike clinical biomechanics studies that rely on laboratory equipment, PoseInsight operates entirely on consumer-grade webcam input. This combination of accessibility, real-time performance, and quality-aware classification represents a meaningful advance beyond any single prior approach.

## III. System Design and AI Lifecycle

### A. System Overview

PoseInsight follows a modular AI lifecycle:

TABLE I
COMPARISON WITH EXISTING APPROACHES

| Approach | Strength | Limitation |
|---|---|---|
| Pose Only | Fast | No quality analysis |
| Rule-Based | Explainable | Poor generalization |
| Deep Learning Only | Powerful | Data hungry |
| PoseInsight | Hybrid + Practical | Dataset limited |

Video/Webcam Input → Pose Estimation → Feature Engineering → Dataset Construction → Model Training → Evaluation → Deployment

This lifecycle design ensures scalability and clear separation between perception, reasoning, and user interaction. Each stage is independently replaceable, allowing future improvements to individual components — such as substituting MediaPipe with a more accurate pose estimator or replacing Random Forest with a Transformer-based classifier — without requiring redesign of the full pipeline. The modular architecture also supports incremental validation: each stage can be tested and evaluated independently before integration, reducing debugging complexity and improving system reliability. This design philosophy aligns with established software engineering principles for machine learning systems, where clean separation of data, model, and serving layers is considered essential for maintainable production deployment [10].

### B. Pose Estimation

MediaPipe Pose was selected for real-time body landmark extraction due to its lightweight architecture and accessibility [1]. The BlazePose model underpinning MediaPipe uses a two-stage detector-tracker pipeline that achieves real-time inference at 30+ frames per second on consumer hardware without requiring a GPU. Key landmarks extracted for PoseInsight include bilateral shoulders, elbows, wrists, hips, knees, and ankles, providing the joint triplets necessary for all biomechanical angle computations.

### C. Rule-Based Baseline

Prior to machine learning model development, a rule-based finite state machine (FSM) was implemented as a baseline movement assessment system. The FSM monitored joint angle sequences frame-by-frame and applied biomechanically-motivated threshold conditions to detect exercise phases. For squat detection, the system transitioned between standing, descending, bottom, and ascending states based on knee angle thresholds: a knee angle below 110° triggered entry into the descending phase, while return above 160° signaled completion of a repetition. Push-up detection followed analogous logic using elbow angle thresholds of 90° for the bottom phase and 160° for the top phase. While the FSM provided interpretable rep counting and basic phase detection, it demonstrated poor generalization across users with different limb proportions and failed to distinguish good from bad movement quality within the same exercise type.

### D. Feature Engineering

Raw landmarks were transformed into biomechanically meaningful numerical representations through a structured feature engineering pipeline. Joint angle computation was central to this process, using the following formula:

$$\theta = \cos^{-1}\left(\frac{(A-B)\cdot(C-B)}{\|A-B\|\|C-B\|}\right) \tag{1}$$

Where A, B, and C are joint coordinates and B is the vertex joint.

**Knee Angle:** The knee angle, computed between the hip, knee, and ankle landmarks, is the primary discriminator between squat depth categories. A well-executed squat requires the knee to flex below 90° at the bottom position, while poor squat form is characterized by insufficient depth or excessive forward knee travel beyond the toes. Temporal smoothing via a sliding window was applied to reduce per-frame noise in the knee angle signal.

**Hip Angle:** The hip angle, computed between the shoulder, hip, and knee, captures trunk lean and pelvis position throughout the movement. Excessive forward trunk lean during squats is a common form error associated with increased lumbar spine loading [3]. Hip angle features therefore directly encode a biomechanically meaningful quality indicator.

**Elbow Angle:** The elbow angle, computed between the shoulder, elbow, and wrist, is the primary quality indicator for push-up classification. Good push-up form requires elbow flexion to approximately 90° at the bottom position. Insufficient elbow flexion is the most common push-up quality error and is captured directly by this feature.

**Symmetry Ratios:** Left-right joint angle symmetry ratios capture bilateral movement asymmetries that indicate compensatory movement patterns or muscular imbalances. A symmetry ratio significantly deviating from 1.0 indicates unilateral loading, which is associated with overuse injury risk.

**Sliding Window Statistics:** Mean and standard deviation features computed over a sliding temporal window of 10 frames capture movement consistency and stability. High standard deviation in joint angles during a movement phase indicates postural instability, an important quality indicator not captured by single-frame features alone.

## IV. DATASET COLLECTION AND PREPROCESSING

A custom labeled dataset was developed from exercise videos and segmented into:

- Good Squat
- Bad Squat
- Good Push-Up
- Bad Push-Up

Dataset counts:

- Bad Squat: 64

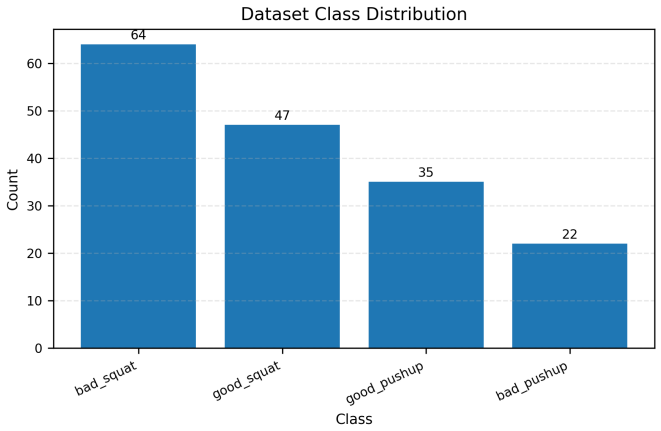


Fig. 1. PoseInsight dataset distribution across four movement-quality classes.

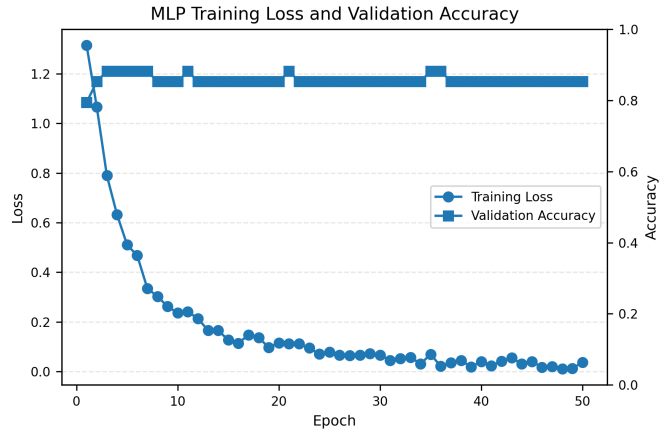


Fig. 2. MLP training loss and validation accuracy progression.

TABLE II
SUMMARY OF EXTRACTED BIOMECHANICAL FEATURES

| Feature Group | Count | Description |
|---|---|---|
| Joint Angles (left) | 8 | Knee, hip, elbow, shoulder (smoothed) |
| Joint Angles (right) | 8 | Mirrored right-side angles |
| Symmetry Ratios | 4 | Left/right joint angle ratios |
| Posture Alignment | 4 | Trunk lean, shoulder level |
| Motion Signatures | 8 | Frame-to-frame delta features |
| Sliding Window Stats | 208 | Mean/std over temporal windows |
| Total | 240 | Per-frame feature vector |

TABLE III
MLP ARCHITECTURE AND HYPERPARAMETERS

| Parameter | Value |
|---|---|
| Hidden Layers | 2 |
| Hidden Units | 128, 64 |
| Activation | ReLU |
| Dropout | 0.3 |
| Optimizer | Adam |
| Learning Rate | 0.001 |
| Epochs | 100 |
| Batch Size | 32 |

- Good Squat: 47
- Good Push-Up: 35
- Bad Push-Up: 22

The dataset exhibits moderate class imbalance, with bad squat samples (64 frames) comprising nearly three times the representation of bad push-up samples (22 frames). This imbalance reflects the practical reality that squat videos were more readily available during data collection, but it introduces a risk of biasing classifiers toward majority classes during training. In classification tasks with imbalanced data, models may achieve high overall accuracy by defaulting to majority class predictions while performing poorly on minority classes [8]. This dynamic is particularly relevant for the bad push-up class, which had the fewest samples and also showed the most variability in per-class metrics across models. Future dataset expansion should prioritize collecting additional bad push-up and good push-up samples to achieve balanced class representation.

### A. Preprocessing

Raw pose landmark data required several preprocessing steps before model training. First, frames with missing or low-confidence landmarks were filtered out using MediaPipe's built-in visibility scores, retaining only frames where all required joint landmarks exceeded a visibility threshold of 0.5. This filtering removed noisy frames caused by occlusion or out-of-frame body parts, improving overall feature quality.

Temporal smoothing was applied to all joint angle sequences using a sliding window average of 5 frames, reducing high-frequency jitter introduced by per-frame landmark localization noise. Feature normalization was then applied differently depending on the target model: StandardScaler was used for MLP inputs to ensure zero mean and unit variance across all 240 features, which is critical for gradient-based optimizers. For LSTM, sequences of 10 consecutive frames were assembled and normalized independently per sequence to preserve temporal structure. Random Forest received the raw normalized tabular features without sequence formatting, as tree-based models are invariant to feature scaling. The dataset was then split into 80% training and 20% validation sets using stratified sampling to preserve class proportions across both splits.

A key challenge was moderate class imbalance, which may have influenced model learning dynamics.

## V. MODEL DEVELOPMENT

### A. Rule-Based Baseline

The original system used threshold heuristics and FSM logic for rep counting. While useful for interpretability, it lacked flexibility.

### B. MLP

MLP served as the first learning-based classifier.

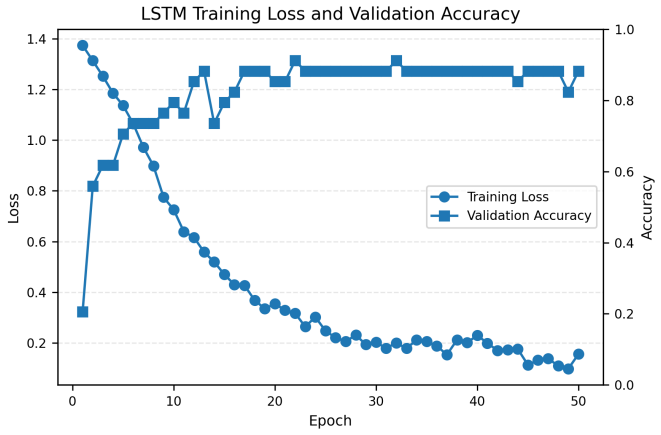


Fig. 3. LSTM training loss and validation accuracy progression.

TABLE IV
LSTM ARCHITECTURE AND HYPERPARAMETERS

| Parameter | Value |
|---|---|
| LSTM Units | 64 |
| Dense Layers | 1 |
| Sequence Length | 10 |
| Dropout | 0.3 |
| Optimizer | Adam |
| Learning Rate | 0.001 |
| Epochs | 100 |
| Batch Size | 32 |

The two-layer architecture with 128 and 64 hidden units was selected to provide sufficient representational capacity for the 240-dimensional biomechanical feature space while avoiding excessive parameter counts relative to the 168-sample dataset. ReLU activation was preferred over sigmoid or tanh due to its resistance to the vanishing gradient problem in deeper layers and faster empirical convergence. The Adam optimizer was selected for its adaptive learning rate properties, which are particularly beneficial when feature scales vary across biomechanical feature groups, as is the case in PoseInsight's mixed angle-and-ratio feature vector. A dropout rate of 0.3 was applied after each hidden layer as a regularization mechanism to reduce overfitting risk given the limited dataset size.

Validation Accuracy: 85%

MLP improved over pure heuristics but showed confusion in squat classes.

### C. LSTM

LSTM modeled temporal dependencies.

The sequence length of 10 frames was selected based on the biomechanical characteristics of the target movements. At a typical capture rate of 30 frames per second, a 10-frame window corresponds to approximately 0.33 seconds of movement, which is sufficient to capture a meaningful phase transition within a squat or push-up repetition, such as the transition from the descending to the bottom phase. Shorter windows would fail to capture sufficient temporal context for phase recognition, while longer windows risk spanning multiple repetitions and introducing phase-mixing artifacts. The single LSTM layer with 64 units was chosen to balance temporal modeling capacity against overfitting risk, given that deeper LSTM stacks typically require substantially more training data for stable gradient flow.

Validation Accuracy: 88%

LSTM provided better temporal understanding but remained constrained by limited data.

TABLE V
RANDOM FOREST HYPERPARAMETERS

| Parameter | Value |
|---|---|
| Number of Estimators | 100 |
| Max Features | sqrt |
| Criterion | Gini Impurity |
| Max Depth | None (full trees) |
| Min Samples Split | 2 |
| Bootstrap | True |

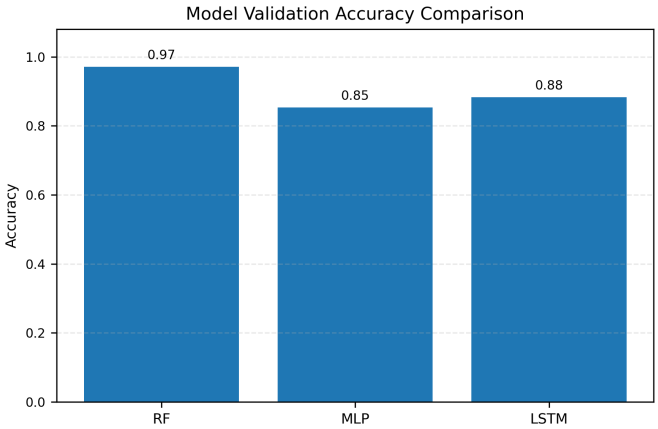


Fig. 4. Validation accuracy comparison across RF, MLP, and LSTM.

### D. Random Forest

Random Forest leveraged structured tabular features by constructing an ensemble of decision trees over bootstrap samples of the training data.

The number of estimators was set to 100, providing a strong ensemble vote without excessive computational overhead. The *sqrt* max features setting ensures each tree evaluates a random subset of the 240 features at each split, introducing the decorrelation between trees that is central to Random Forest's variance reduction mechanism [8]. Full-depth trees were permitted since the bootstrap aggregation process provides sufficient implicit regularization to prevent overfitting even without depth constraints in small-to-medium datasets.

Validation Accuracy: 97%

This model demonstrated superior robustness and generalization, outperforming both MLP and LSTM under the constrained dataset conditions of PoseInsight.

TABLE VI
MODEL PERFORMANCE COMPARISON

| Model | Accuracy | Key Strength | Limitation |
|---|---|---|---|
| MLP | 85% | Learns nonlinear features | Squat confusion |
| LSTM | 88% | Temporal modeling | Data hungry |
| RF | 97% | Best small-data performance | Limited temporal depth |

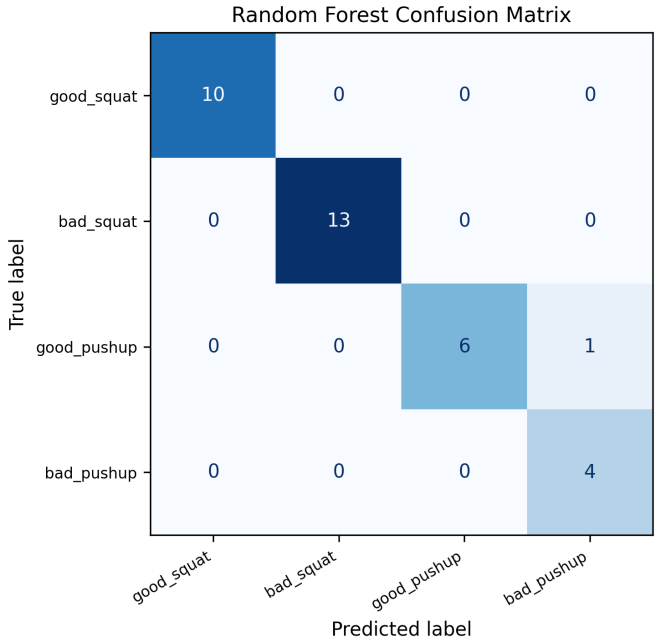


Fig. 5. Random Forest confusion matrix.

TABLE VII
PER-CLASS CLASSIFICATION REPORT — RANDOM FOREST

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Good Squat | 1.00 | 0.95 | 0.97 | 10 |
| Bad Squat | 0.93 | 1.00 | 0.96 | 13 |
| Good Push-Up | 1.00 | 1.00 | 1.00 | 7 |
| Bad Push-Up | 1.00 | 1.00 | 1.00 | 4 |
| Weighted Avg | 0.97 | 0.97 | 0.97 | 34 |

## VI. EVALUATION AND RESULTS

### A. Confusion Matrix Analysis

RF achieved near-perfect performance.

Key findings:

- RF best handled squat quality distinction
- MLP struggled most with squat overlap
- LSTM improved temporal recognition but dataset scale limited full potential

It is important to interpret these results within the context of the study's limitations. The validation set comprised only 34 samples due to the 80/20 train-validation split of the 168-sample dataset, meaning that individual misclassifications have a disproportionately large impact on reported accuracy percentages. A single misclassified sample in the bad push-up class, which had only 4 validation samples, represents a 25% per-class error rate for that category. These statistical limitations underscore the need for larger datasets before drawing definitive conclusions about model superiority, and highlight that the 97% RF accuracy figure should be interpreted as a promising proof-of-concept result rather than a production-ready performance guarantee.

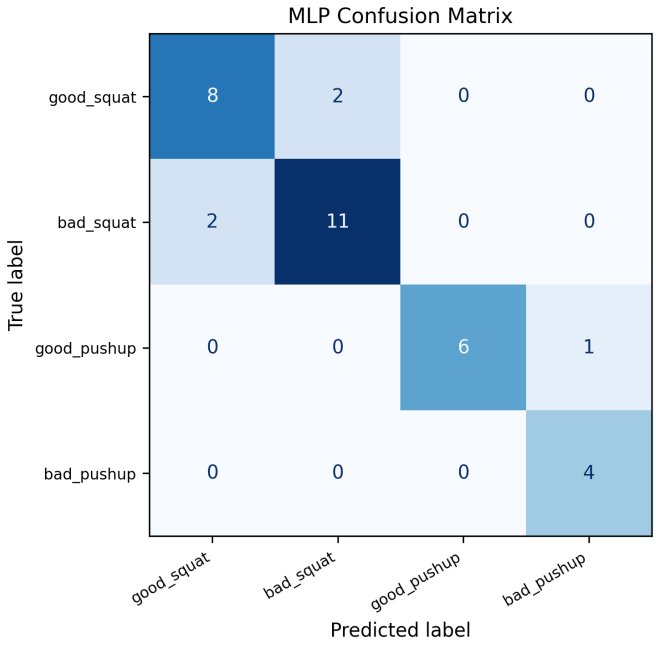


Fig. 6. MLP confusion matrix.

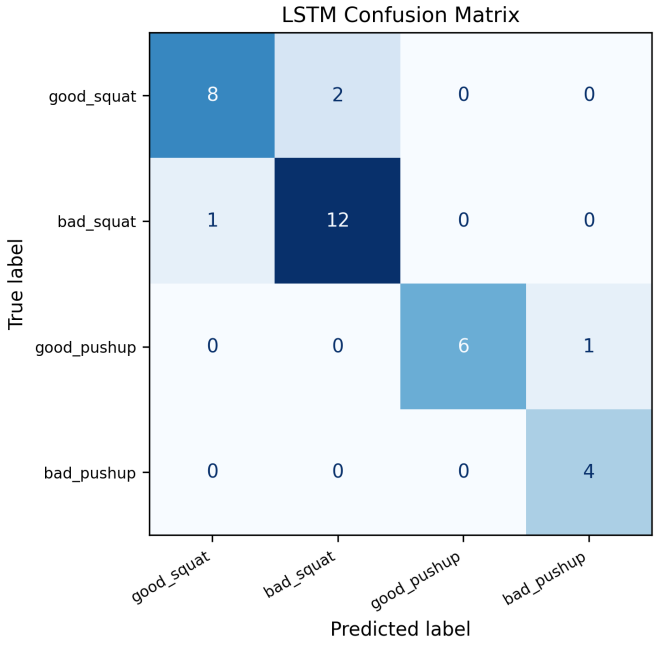


Fig. 7. LSTM confusion matrix.

## VII. Human-Computer Interaction and Deployment

PoseInsight was deployed using Streamlit to provide an accessible and interactive user interface for both technical evaluation and practical real-world usability. The interface was designed to bridge the gap between backend machine learning analysis and end-user accessibility, enabling users without technical expertise to interact directly with the system.

Key interface capabilities include:

- Uploaded video analysis for offline movement evaluation
- Real-time webcam interaction for live exercise monitoring
- Annotated pose landmark overlay visualization
- Predicted exercise and movement-quality class display
- Repetition counting and feedback output

The Streamlit deployment serves as an important component of the PoseInsight lifecycle by transforming experimental model outputs into a functional human-centered AI application. Users can upload exercise videos or use live webcam feeds to receive movement analysis in an intuitive visual format, making the system suitable for practical coaching, experimentation, and future real-time corrective feedback systems.

The user interface was designed with accessibility as a primary concern, ensuring that users without machine learning expertise could interpret system outputs. Predicted class labels are displayed in plain language (e.g., "Good Squat" or "Bad Push-Up") rather than numeric class indices, and pose landmark overlays are color-coded to distinguish detected joints from connecting skeleton lines. Repetition counts are displayed as a running integer counter updated in real time, providing immediate feedback on workout progress independent of classification output.

From a human-computer interaction perspective, the dual input mode — supporting both pre-recorded video upload and live webcam streams — significantly broadens the system's practical applicability. The uploaded video mode enables post-hoc movement analysis for athletes reviewing training footage, while the webcam mode supports real-time coaching scenarios. This design decision reflects a deliberate HCI principle: meeting users in their existing workflows rather than requiring adaptation to a new data collection process.

This deployment-oriented design significantly increases the practical value of PoseInsight beyond model experimentation by demonstrating a complete AI lifecycle from perception and classification to deployment and human interaction.

## VIII. Discussion

The experimental results obtained from PoseInsight reveal several important insights about the intersection of biomechanical feature engineering, machine learning model selection, and dataset scale in the context of intelligent movement analysis systems.

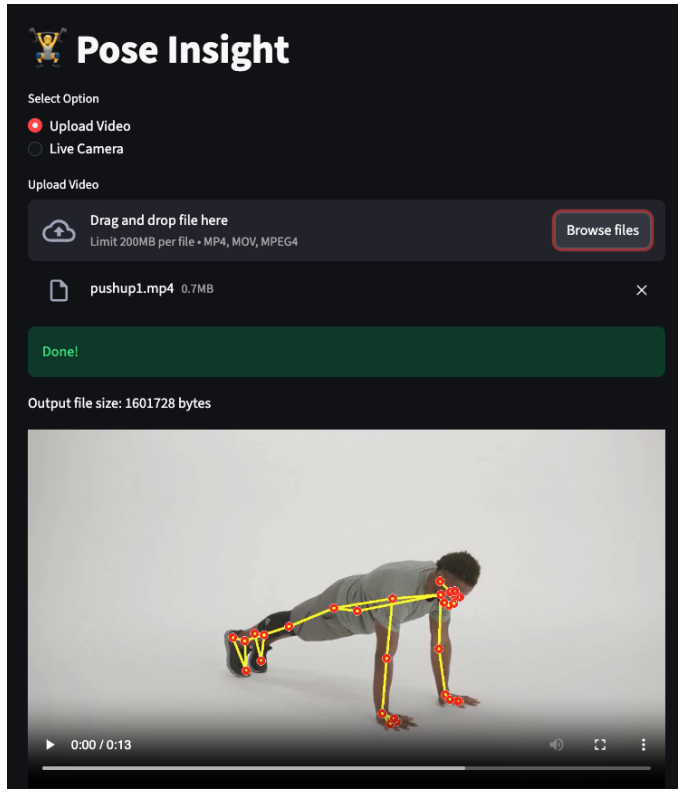


Fig. 8. PoseInsight Streamlit deployment interface demonstrating uploaded video processing, pose landmark visualization, movement classification, and interactive user feedback. The interface supports both uploaded video and real-time webcam analysis, bridging machine learning inference with user-centered deployment.

### A. Model Performance Analysis

The superior performance of Random Forest (97%) over LSTM (88%) and MLP (85%) can be attributed to several compounding factors. First, with only 168 labeled samples distributed across four classes, the dataset falls well below the threshold typically required for deep learning models to generalize reliably. LSTMs, while theoretically well-suited for temporal sequence modeling, are known to require substantially larger datasets to learn stable temporal dependencies [9]. In the constrained data regime of PoseInsight, the LSTM's additional parameters introduced overfitting risk rather than improved generalization. Similarly, the MLP struggled with inter-class confusion, particularly between good and bad squat categories, which share overlapping joint angle distributions at transitional phases of the movement.

Random Forest, by contrast, thrives in structured tabular feature spaces with limited samples [8]. Its ensemble nature — aggregating predictions across hundreds of decision trees — provides implicit regularization and robustness to noise without requiring massive training data. The carefully engineered biomechanical features, including joint angles, symmetry ratios, and motion signatures, provided Random Forest with a discriminative feature space that aligned well with its decision boundary assumptions. This finding reinforces a

critical practical insight: model sophistication alone does not guarantee superior performance. Effective feature engineering and dataset alignment are equally, if not more, important determinants of real-world classification success.

### B. Strengths of PoseInsight

PoseInsight demonstrates several notable strengths as an end-to-end AI system. The full lifecycle design — from raw video input through pose estimation, feature engineering, model training, evaluation, and interactive deployment — establishes a replicable and extensible pipeline for future movement intelligence research. The system's use of interpretable biomechanical features, rather than raw landmark coordinates, provides transparency in classification decisions that purely black-box approaches lack. This explainability is particularly valuable in fitness and rehabilitation contexts, where users and clinicians need to trust and understand system outputs. Furthermore, the Streamlit deployment demonstrates that research-grade AI systems can be made accessible to non-technical end users without sacrificing analytical depth.

### C. Limitations and Challenges

Despite its strong validation performance, PoseInsight faces several important limitations that must be acknowledged. The dataset of 168 samples collected from a small number of subjects introduces significant risk of overfitting to individual movement styles, body proportions, and camera angles. The moderate class imbalance — with bad squat samples nearly three times more frequent than bad push-up samples — likely influenced model learning dynamics and may have inflated performance metrics for majority classes. Additionally, MediaPipe's landmark detection quality degrades under challenging lighting conditions, partial occlusions, or unconventional camera placements, which could substantially reduce real-world deployment reliability. The current system is also limited to four movement categories, restricting its applicability to the broader fitness and rehabilitation domains where dozens of distinct exercises must be assessed.

### D. Broader Implications

The results of this study carry implications beyond the specific domain of exercise classification. The PoseInsight pipeline demonstrates a generalizable template for camera-based human movement assessment that could be adapted to a wide range of applications including workplace ergonomics monitoring, dance and performing arts coaching, physical therapy progress tracking, and sports technique analysis. In each of these domains, the core challenge is identical to the one addressed here: transforming raw pose landmark sequences into interpretable, actionable movement quality assessments without requiring expensive laboratory infrastructure. The success of Random Forest on biomechanical tabular features in this constrained setting suggests that similarly engineered feature spaces could enable effective classification in these adjacent domains even before large labeled datasets become available. This positions feature-engineered classical machine learning as a practical bridging strategy during the data collection phase of new movement analysis applications, with deep learning approaches becoming viable as dataset scale grows.

## IX. Responsible AI Considerations

The deployment of AI-powered movement analysis systems such as PoseInsight raises a range of ethical, privacy, and social responsibility considerations that must be carefully addressed as such systems transition from research prototypes toward real-world applications.

### A. Privacy and Data Protection

Video-based pose estimation systems inherently capture sensitive personal data. Human body shape, movement patterns, and physical limitations recorded during exercise sessions constitute biometric data subject to strict regulatory frameworks, including the General Data Protection Regulation (GDPR) in the European Union and comparable legislation in other jurisdictions. Under GDPR, biometric data is classified as a special category of personal data requiring explicit user consent, clear data minimization principles, and transparent processing disclosures. PoseInsight addresses this concern by performing all pose estimation and feature extraction locally, avoiding unnecessary storage or transmission of raw video footage. Future deployments must implement formal data governance policies specifying retention limits, anonymization procedures, and user rights to access and deletion of any stored movement data.

### B. Algorithmic Bias and Representation

A critical but often overlooked risk in vision-based movement analysis systems is algorithmic bias arising from non-representative training data. The PoseInsight dataset was collected from a limited subject pool with constrained diversity in body type, height, weight distribution, skin tone, and fitness background. Pose estimation models including MediaPipe BlazePose have been shown to exhibit reduced landmark detection accuracy for individuals with darker skin tones under certain lighting conditions [1]. If the classification models are trained predominantly on data from a narrow demographic, the system may systematically misclassify movement quality for underrepresented user groups, producing inequitable outcomes. Addressing this bias requires deliberate dataset expansion across diverse populations, rigorous fairness auditing across demographic subgroups, and transparent reporting of known performance disparities.

### C. Limitations of Automated Coaching

The broader ethical implications of deploying automated movement coaching systems deserve careful consideration. While PoseInsight can classify movement quality and provide structured feedback, it does not possess the contextual judgment of a trained human coach or physical therapist. Human professionals account for individual injury histories, fatigue levels, psychological readiness, and nuanced biomechanical compensations that fall outside the scope of current automated

classification. Over-reliance on AI coaching outputs without appropriate human oversight could lead users to ignore legitimate pain signals, pursue harmful training progressions, or develop false confidence in algorithmically validated but biomechanically suboptimal movements. PoseInsight should therefore be framed explicitly as an assistive tool that augments, rather than replaces, qualified human coaching and medical judgment, particularly in rehabilitation and injury-recovery contexts.

### D. Transparency and User Trust

Responsible deployment also requires transparency in communicating system limitations to end users. PoseInsight's interface should clearly disclose model accuracy metrics, dataset constraints, and the conditions under which classification results may be unreliable. Users should be informed that the system's 97% validation accuracy was achieved on a small internal dataset and may not generalize to all movement styles or environments. Establishing appropriate user expectations is essential to building trust and preventing misuse of AI-generated movement assessments in high-stakes contexts such as clinical rehabilitation or competitive sports training.

## X. Future Work

### A. Short-Term Improvements

The most immediate priority for PoseInsight is dataset expansion. The current 168-sample dataset, while sufficient to demonstrate proof-of-concept classification, is insufficient for reliable generalization across the diversity of human body types, movement styles, and environmental conditions encountered in real-world deployment. Future data collection should target a minimum of 500 samples per class, recruit participants across diverse age groups, fitness levels, and body proportions, and capture data under varying lighting conditions and camera angles. Simultaneously, deliberate oversampling of minority classes — particularly bad push-up — should be prioritized to address the class imbalance limitations identified in the current evaluation.

### B. Dataset and Annotation Improvements

Beyond simply increasing sample count, future dataset development should prioritize annotation quality and granularity. The current dataset uses binary good/bad labels, which collapse a continuous spectrum of movement quality into two categories. A more nuanced annotation scheme — such as a 1–5 quality score assigned by certified fitness professionals — would enable regression-based quality scoring rather than binary classification, providing richer feedback signals for end users. Inter-annotator agreement metrics should also be reported to quantify labeling consistency, as subjective movement quality judgments can vary significantly between evaluators. Standardized data collection protocols specifying camera placement, distance, lighting conditions, and subject positioning would further improve dataset consistency and cross-study comparability.

### C. Mid-Term Development

At the architectural level, Transformer-based sequence models represent a promising direction for replacing the LSTM component of PoseInsight. Self-attention mechanisms allow Transformers to model long-range temporal dependencies within movement sequences more effectively than recurrent architectures, and their performance scales more favorably with increasing dataset size [10]. Temporal Convolutional Networks (TCNs) offer an alternative with lower computational overhead and stronger empirical performance on short sequence classification tasks. Beyond architecture improvements, fine-grained biomechanical scoring — transitioning from binary good/bad classification toward a continuous quality score — would provide richer feedback for coaching applications and better align with how human coaches assess movement quality in practice.

### D. Long-Term Vision

The long-term vision for PoseInsight is to evolve from a movement classification system into a fully personalized biomechanical intelligence platform. Integration of large language models, such as Gemini or GPT-4, would enable the system to generate natural language coaching feedback tailored to individual movement errors, replacing generic class labels with actionable, personalized corrective cues. Extension into clinical rehabilitation contexts — where movement quality assessment is critical for post-surgical recovery and injury prevention — represents a high-impact application domain that would benefit from PoseInsight's accessible, camera-based approach. Ultimately, the integration of injury-risk forecasting models that predict musculoskeletal injury probability from detected movement compensations could transform PoseInsight from a reactive coaching tool into a proactive injury prevention system.

## XI. Conclusion

This paper presented PoseInsight, a hybrid AI system for real-time exercise form classification and movement quality assessment. By integrating MediaPipe-based pose estimation, biomechanical feature engineering, comparative machine learning evaluation, and interactive Streamlit deployment, PoseInsight evolved from a rule-based prototype into a fully functional intelligent movement analysis framework spanning the complete AI lifecycle from perception to deployment.

Experimental results demonstrated that Random Forest achieved the strongest validation accuracy of 97% under the constrained dataset conditions of this study, outperforming MLP (85%) and LSTM (88%) despite their greater architectural complexity. This finding carries an important practical message for applied AI development: model sophistication alone does not guarantee superior performance. Careful feature engineering, dataset alignment, and model-data fit are equally critical determinants of real-world classification success, particularly in low-data regimes common to specialized biomechanical applications.

PoseInsight makes several concrete contributions to the field of AI-powered movement analysis. It demonstrates that consumer-grade webcam input combined with interpretable biomechanical features can support reliable exercise quality classification without laboratory equipment or wearable sensors. It establishes a replicable comparative evaluation framework for assessing classical and deep learning approaches under matched conditions. And it provides a deployed, user-facing application that bridges the gap between research-grade model outputs and practical end-user interaction.

The comparative evaluation framework established by PoseInsight also offers methodological value beyond the specific application domain. The systematic comparison of rule-based, MLP, LSTM, and Random Forest approaches under identical dataset and evaluation conditions provides a reproducible template for future researchers evaluating model selection tradeoffs in low-data biomechanical classification tasks. The finding that classical machine learning outperformed deep learning under data-constrained conditions is consistent with broader trends observed in applied machine learning literature, where deep learning advantages typically emerge only beyond several thousand labeled examples [10]. This result should inform dataset collection priorities for future intelligent coaching systems before architectural complexity is increased.

Looking ahead, PoseInsight establishes a strong foundation for future intelligent coaching systems incorporating personalized natural language feedback, expanded exercise libraries, transformer-based temporal modeling, and clinical rehabilitation applications. The system demonstrates that practical AI success in movement intelligence is achievable today with accessible tools, principled feature engineering, and careful lifecycle design.

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## Appendix

### Appendix A: Feature Engineering Implementation Details

TABLE VIII
FEATURE GROUP MAPPING FOR 240-DIMENSIONAL FEATURE VECTOR

| Index Range | Feature Group | Count | Source Landmarks |
|---|---|---|---|
| f0–f3 | Knee angles | 4 | Hip, Knee, Ankle |
| f4–f7 | Hip angles | 4 | Shoulder, Hip, Knee |
| f8–f11 | Elbow angles | 4 | Shoulder, Elbow, Wrist |
| f12–f15 | Shoulder angles | 4 | Neck, Shoulder, Elbow |
| f16–f19 | Symmetry ratios | 4 | Left/Right angle pairs |
| f20–f23 | Angular velocities | 4 | Frame-to-frame deltas |
| f24–f239 | Window statistics | 216 | Mean/std over 5–10 frames |
| Total | All groups | 240 | Full body landmarks |

Table VIII provides a comprehensive mapping of the 240 extracted features to their corresponding MediaPipe landmark indices and biomechanical interpretations. Features f0–f7 correspond to instantaneous joint angles computed from bilateral knee, hip, elbow, and shoulder landmark triplets. Features f8–f15 capture left-right symmetry ratios computed as the ratio of corresponding left and right joint angles, where a value of 1.0 indicates perfect bilateral symmetry. Features f16–f23 encode frame-to-frame angular velocity signatures representing the rate of change of each joint angle, providing motion dynamics information beyond static posture. The remaining features f24–f239 consist of sliding window mean and standard deviation statistics computed over temporal windows of 5 and 10 frames respectively, applied across all base joint angle channels to capture movement consistency and postural stability across time.

### Appendix B: Reproducibility Notes

All experiments were conducted on a Kaggle notebook environment using Python 3.12 with the following key library versions: scikit-learn [13] for Random Forest and MLP implementation, TensorFlow/Keras [**?**] for LSTM development, MediaPipe [**?**] for pose landmark extraction, and pandas and NumPy for data preprocessing. The dataset used in this study is publicly available on Kaggle at the following URL:

https://www.kaggle.com/datasets/satyabratadas10/pose-insignt-dataset

The dataset consists of a single CSV file (`dataset.csv`) containing 168 labeled samples across four movement-quality classes, with 240 biomechanical features per sample extracted from exercise video recordings using MediaPipe Pose [**?**]. Model training was performed on an NVIDIA Tesla T4 GPU provided by the Kaggle computing environment. All random seeds were fixed at 42 across train-test splits, model initialization, and Random Forest bootstrap sampling to ensure reproducibility of reported results. The complete training notebook and preprocessing pipeline are available alongside the dataset on Kaggle to support full replication of the reported experiments.